KAIZEN MATHS

Lesson starter

Momentum and Impulse

Momentum and Impulse

Name _____

S1

A particle of mass 3 kg moves left at 4 m s^{-1}. Find its momentum.

S2

A constant force of 6 N acts right on a particle for 0.25 s. Find the impulse.

S3

A particle of mass 3 kg is moving to the right at 8 m s^{-1}. After an impulse, its velocity is 8 m s^{-1} to the left. Find the impulse on the particle.

S4

A particle of mass 3 kg changes velocity from 12 m s^{-1} to the right to 9 m s^{-1} to the right in 1 s. Find the average force on the particle.

Show your working

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MEC-503C-L1 · S

Show your working

KAIZEN MATHS

Exit ticket

Momentum and Impulse

Momentum and Impulse

Name _____

E1

A particle of mass 6 kg moves left at 12 m s^{-1}. Find its momentum.

E2

A constant force of 20 N acts right on a particle for 0.2 s. Find the impulse.

E3

A particle of mass 3 kg is moving to the right at 8 m s^{-1}. After an impulse, its velocity is 2 m s^{-1} to the left. Find the impulse on the particle.

KAIZEN MATHS

Exit ticket

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Name _____

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A particle of mass 6 kg moves left at 12 m s^{-1}. Find its momentum.

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A constant force of 20 N acts right on a particle for 0.2 s. Find the impulse.

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A particle of mass 3 kg is moving to the right at 8 m s^{-1}. After an impulse, its velocity is 2 m s^{-1} to the left. Find the impulse on the particle.

MEC-503C-L1 · E

Show your working

KAIZEN MATHS

Exit ticket

Momentum and Impulse

Momentum and Impulse

Name _____

E1

A particle of mass 6 kg moves left at 12 m s^{-1}. Find its momentum.

E2

A constant force of 20 N acts right on a particle for 0.2 s. Find the impulse.

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A particle of mass 3 kg is moving to the right at 8 m s^{-1}. After an impulse, its velocity is 2 m s^{-1} to the left. Find the impulse on the particle.

MEC-503C-L1 · E

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A particle of mass 3 kg is moving to the right at 8 m s^{-1}. After an impulse, its velocity is 2 m s^{-1} to the left. Find the impulse on the particle.

MEC-503C-L1 · E

Show your working

Teacher answers & working

Momentum and Impulse · Momentum and Impulse

MEC-503C-L1 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S1

A particle of mass 3 kg moves left at 4 m s^{-1}. Find its momentum.

Answer:

12 kg m s^{-1} to the left

Working:

Momentum is mass multiplied by velocity.

$p = mv$

Take right as positive. A velocity to the left is negative.

$m = 3, v = -4$

$p = 3 \times (-4)$

$p = -12$ kg m s^{-1}

Momentum = 12 kg m s^{-1} to the left.

S2

A constant force of 6 N acts right on a particle for 0.25 s. Find the impulse.

Answer:

1.5 N s to the right

Working:

Impulse from a constant force is force multiplied by time.

Impulse = Ft

Take right as positive. A force to the left is negative.

$F = 6, t = 0.25$

Impulse = (6) × 0.25

Impulse = 1.5 N s

Impulse = 1.5 N s to the right.

Teacher answers & working

Momentum and Impulse · Momentum and Impulse

MEC-503C-L1 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S3

A particle of mass 3 kg is moving to the right at 8 m s^{-1}. After an impulse, its velocity is 8 m s^{-1} to the left. Find the impulse on the particle.

Answer:

48 N s to the left

Working:

Take right as positive.

$u = 8, v = -8, m = 3$

Impulse equals change in momentum.

Impulse = mv - mu

Impulse = m(v - u)

Impulse = 3(-8 - 8)

Impulse = 3 × -16

Impulse = -48 N s

Impulse = 48 N s to the left.

S4

A particle of mass 3 kg changes velocity from 12 m s^{-1} to the right to 9 m s^{-1} to the right in 1 s. Find the average force on the particle.

Answer:

9 N to the left

Working:

Take right as positive.

$u = 12, v = 9, m = 3, t = 1$

First find the change in momentum.

Impulse = m(v - u)

Impulse = 3(9 - 12) = -9 N s

Average force is impulse divided by time.

$F = -9 \div 1$

$F = -9$ N

Average force = 9 N to the left.

Teacher answers & working

Momentum and Impulse · Momentum and Impulse

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E1

A particle of mass 6 kg moves left at 12 m s^{-1}. Find its momentum.

Answer:

72 kg m s^{-1} to the left

Working:

Momentum is mass multiplied by velocity.

$p = mv$

Take right as positive. A velocity to the left is negative.

$m = 6, v = -12$

$p = 6 \times (-12)$

$p = -72$ kg m s^{-1}

Momentum = 72 kg m s^{-1} to the left.

E2

A constant force of 20 N acts right on a particle for 0.2 s. Find the impulse.

Answer:

4 N s to the right

Working:

Impulse from a constant force is force multiplied by time.

Impulse = Ft

Take right as positive. A force to the left is negative.

$F = 20, t = 0.2$

Impulse = (20) × 0.2

Impulse = 4 N s

Impulse = 4 N s to the right.

Teacher answers & working

Momentum and Impulse · Momentum and Impulse

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E3

A particle of mass 3 kg is moving to the right at 8 m s^{-1}. After an impulse, its velocity is 2 m s^{-1} to the left. Find the impulse on the particle.

Answer:

30 N s to the left

Working:

Take right as positive.

$u = 8, v = -2, m = 3$

Impulse equals change in momentum.

Impulse = mv - mu

Impulse = m(v - u)

Impulse = 3(-2 - 8)

Impulse = 3 × -10

Impulse = -30 N s

Impulse = 30 N s to the left.